

Requirements

You need to implement **two main modules**, following **REST principles**, **Domain-Driven Architecture**, and using **TypeScript** with a **relational database** (like PostgreSQL).

Module 1: AI Chat Module

- Accepts a user question.
- Returns a **mocked OpenAI response**.
- Stores the question, answer and tokens in the database.
- Track monthly usage per user:
 - Each user gets 3 free messages per month.
 - After that, a valid subscription bundle is required.
- Support multiple active bundles per user with different tiers:
 - Basic (10 responses).
 - Pro (100 responses).
 - Enterprise (unlimited).
- Deduct usage from the bundle with the latest remaining quota.
- Auto reset free quota on the 1st of each month.
- Throw structured errors if the user exceeds quota.
- Constraints:
 - Simulate Open Ai Api response time delay (mocked).

Module 2: Subscription Bundle Module

- Allow users to:
 - Create subscription bundles (**Basic**, **Pro**, **Enterprise**)
 - Choose billing cycle: monthly or yearly
 - Toggle auto-renew
- Each subscription includes:
 - maxMessages, price, startDate, endDate, renewalDate
- Simulate billing logic:
 - Auto-renew subscriptions if enabled
 - If "payment fails" (randomly), mark subscription inactive
- Support cancellation:
 - Ends current billing cycle
 - Prevents renewal

- Preserves usage history

Architecture

- Use **Clean Architecture (DDD-style)**
- Layers: **domain/entities**, **services**, **repositories**, **controllers**
- Organize modules independently (e.g., **chat/**, **subscriptions/**)
- TypeScript-first.
- Configure prettier and Eslint.

Submission Guidelines

Please follow the instructions below carefully:

1. **Create a public GitHub repository** to manage your code.
2. **Name the repository using your full name.**
3. **Add a **README.md** file** that clearly explains your code and any other relevant details.
4. **Upload the provided test PDF file** to the repository as part of your submission.
5. Once you've completed the task, **reply to this email with the link to your GitHub repo.**

Deadline: You have **48 hours** to complete this task.

Submissions will not be entertained after the deadline.